15.School Administration Management System

(Using Core Java & OOP Concepts)

Under the Guidance of

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Submitted By

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Object Oriented Programming With Java

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ABSTRACT :

School Administration Management System is a software application designed to manage and organize various activities in a school efficiently. The project aims to reduce manual work and improve data handling using Object-Oriented Programming concepts.

This system helps in managing student records, teacher information, attendance, fee details, and examination results. Traditionally, these tasks are handled manually, which is time-consuming and prone to errors. This system provides a digital solution to store, update, and retrieve data quickly and securely.
The
The application includes different modules such as Student Management, Teacher Management, Attendance, Fee Management, and Result Processing. It ensures better accuracy, saves time, and simplifies administrative tasks.

Overall, the project demonstrates the use of OOP principles such as encapsulation, inheritance, polymorphism, and abstraction while providing a functional and efficient system for school management.

INTRODUCTION:

The School Administration Management System is developed to automate the daily operations of a school. Managing student and teacher data manually is difficult and inefficient. This system provides a structured and digital way to handle all school-related data.

The system includes various features such as adding and managing student details, assigning teachers, recording attendance, managing fees, and generating results. It helps reduce paperwork and improves data accuracy.

Objectives:
- To manage student and teacher records
- To automate attendance and fee tracking
- To reduce manual work
- To implement OOP concepts in a real-world application

SCOPE:

This project is suitable for small to medium-sized schools. It can efficiently manage daily administrative tasks. In the future, it can be extended by integrating a database and developing a graphical user interface (GUI) for better usability.

SYSTEM REQUIREMENTS:

Hardware:
- Computer with at least 4GB RAM

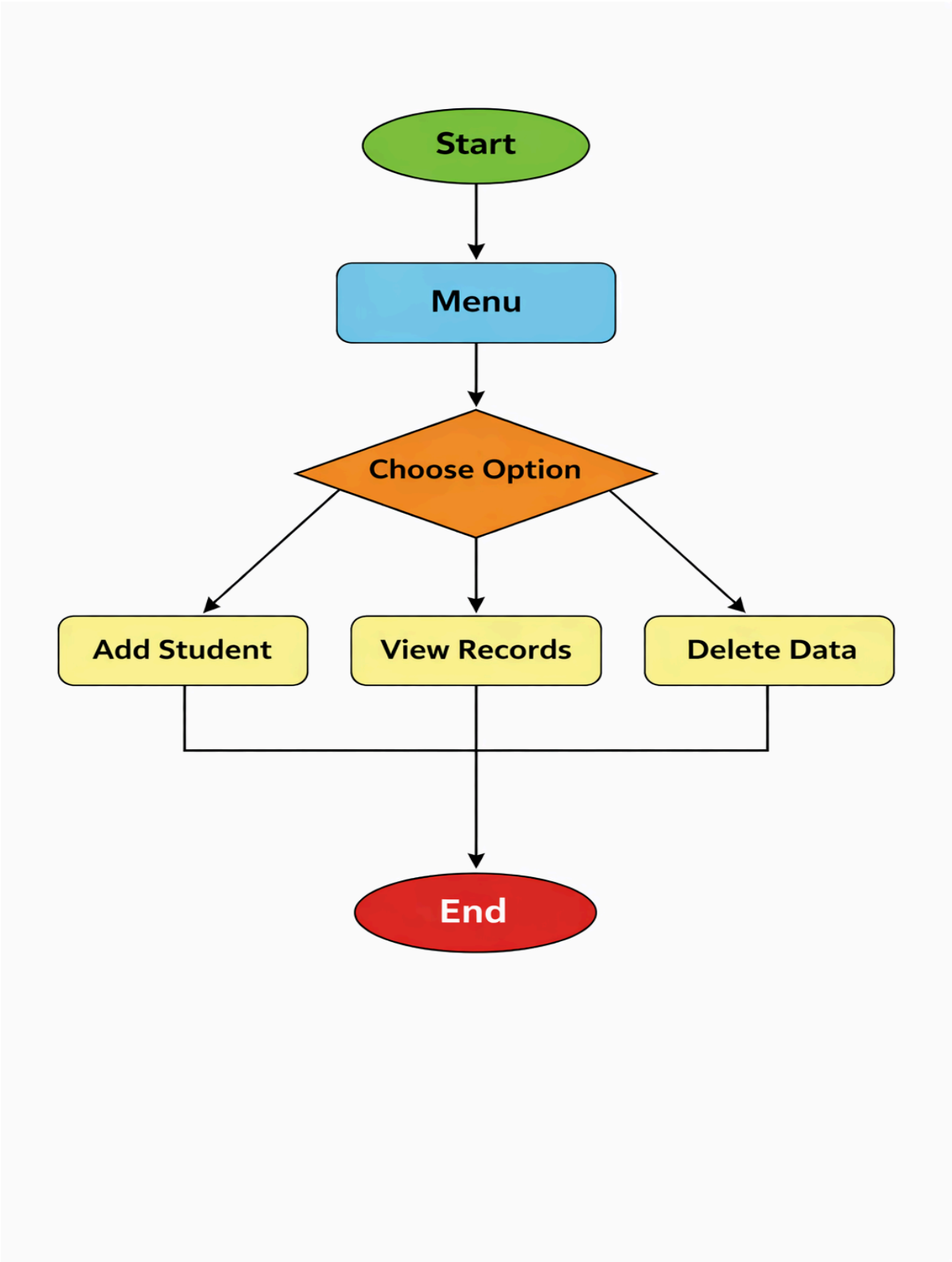
Software:
- Java JDK / C++ Compiler
- IDE (VS Code / Eclipse / CodeBlocks)

METHODOLOGY:

The system works in a step-by-step process. First, the user selects an option from the menu. Based on the choice, the system performs operations such as adding a student, viewing records, updating information, or deleting data.

Each module is handled using separate classes. Data is stored in arrays or lists. The system allows easy navigation through menu-driven options and ensures smooth execution of operations.

FLOWCHART:


Start
Menu
Choose Option
Add Student
View Records
Delete Data
End

IMPLEMENTATION (LOGIC EXPLAIN):

The project consists of multiple classes such as Student, Teacher, Attendance, Fee, and Result.

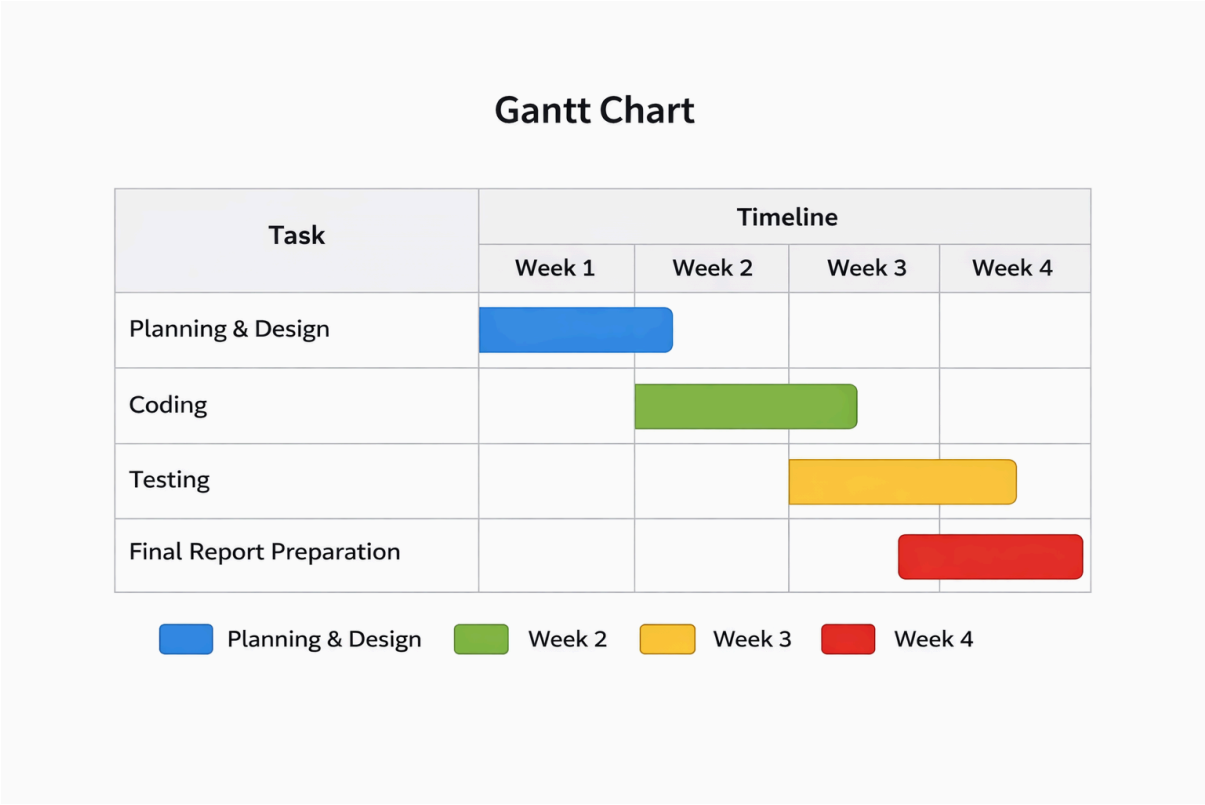
- The Student class stores student details like name, roll number, and class.
- The Teacher class manages teacher information and subjects.
- The Attendance class records daily attendance.
- The Fee class handles fee payment status.
- The Result class stores marks and calculates results.

All classes are designed using OOP concepts. Inheritance is used to derive Student and Teacher from a common Person class. Encapsulation is applied by keeping variables private and using getter/setter methods.

FUTURE ENHANCEMENTS:

- Add database integration (MySQL)
- Develop GUI using Java Swing
- Add login authentication system
- Online access for students and teachers

GANTT CHART:



CONCLUSION:

The School Administration Management System successfully demonstrates the use of Object-Oriented Programming concepts. It provides an efficient way to manage school operations and reduces manual effort. The system is user-friendly and can be further enhanced with advanced features.

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[2] Oracle, "Java Documentation." [Online]. Available: https://docs.oracle.com

[3] S. Kumar, "AI in Education," Proc. IEEE Conf., 2021.